

Further Statistics 1

Notes · Edexcel A Level Further Mathematics — Further Statistics 1 · Chapter 2: Poisson Distribution

Chapter 2: Poisson Distribution

Why this chapter matters

The Poisson distribution is your first genuinely new named distribution in Further Statistics 1. It models the number of times a random event happens in a fixed interval of time or space — phone calls arriving at a call centre, typos on a page, radioactive decays per minute — whenever those events happen independently and at a constant average rate. It also turns up as a very useful *approximation* to the Binomial distribution when n is large and p is small, which is a standard exam trick worth mastering.

Concept: conditions for a Poisson model

A random variable X follows a Poisson distribution, written $X \sim \text{Po}(\lambda)$, when:

1. Events occur singly, at random, and independently of one another.
2. Events occur at a constant average rate λ over the given interval.
3. The interval can be time, length, area, or volume.

λ (lambda) is both the mean number of occurrences in the interval AND, remarkably, the variance — a fact you should always quote and never derive under exam time pressure.

Key formula: Poisson probability, mean and variance

KEY FORMULA

Quantity	Formula
$P(X = x)$	$e^{-\lambda} \lambda^x / x!$ for $x = 0, 1, 2, 3, \dots$
$E(X)$	λ
$\text{Var}(X)$	λ

Because $E(X) = \text{Var}(X) = \lambda$, a quick check on whether Poisson is a sensible model for real data is to see whether the sample mean and sample variance are roughly equal. Note $\text{SD}(X) = \sqrt{\lambda}$, not λ — a frequent slip when a question specifically asks for a standard deviation rather than a variance.

Worked example 1: basic Poisson probability

EXAMPLE 2.1

Question. The number of emails Priya receives per hour follows a Poisson distribution with mean 3.5. Find the probability that in a randomly chosen hour she receives (a) exactly 2 emails, (b) at least 1 email.

Step 1. State the model: $X \sim \text{Po}(3.5)$. **(B1)**

Step 2. (a) $P(X = 2) = e^{-3.5}(3.5)^2/2! = e^{-3.5}(12.25)/2 \approx 0.1850$. **(M1A1)**

Step 3. (b) Use the complement: $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-3.5} \approx 1 - 0.0302 = 0.9698$. **(M1A1)**

Answer: (a) 0.185 (3 s.f.), (b) 0.970 (3 s.f.)

EXAM TIP

Make “at least 1” an automatic complement reflex — trying to sum $P(X=1) + P(X=2) + \dots$ to infinity waste huge amounts of time and usually go wrong. Also set up your calculator to evaluate $e^{-\lambda}$ directly (it is common to round λ needlessly first, losing accuracy marks).

Concept: sum of independent Poisson variables

If $X \sim \text{Po}(\lambda)$ and $Y \sim \text{Po}(\mu)$ are **independent**, then $X + Y \sim \text{Po}(\lambda + \mu)$. This extends to any number of independent Poisson variables: their sum is Poisson with parameter equal to the sum of the individual means. This is one of the most exam-friendly facts in the whole unit because it requires no calculation to justify — just correct identification.

Worked example 2: sum of independent Poisson variables

EXAMPLE 2.2

Question. Faults on a length of cable occur as a Poisson process at a rate of 0.4 per metre. Find the probability that a randomly chosen 10 m length of cable has exactly 3 faults.

Step 1. Let X be the number of faults in 10 m. Since faults occur at a constant rate independently along the cable, X is the sum of ten independent $\text{Po}(0.4)$ variables for each metre, so $X \sim \text{Po}(10 \times 0.4) = \text{Po}(4)$. **(M1)**

Step 2. $P(X = 3) = e^{-4}(4^3)/3! = e^{-4}(64)/6 \approx 0.1954$. **(M1A1)**

Answer: 0.195 (3 s.f.)

EXAM TIP

This “scaling up the interval” idea (rate per metre $\rightarrow$ rate for 10 metres) is really just the sum-of-independent-Poissons result in disguise, and it is tested constantly. Understand the link rather than learning “multiply λ by the new length” as an isolated rule — if you understand why it works, you can also handle combined contexts, e.g. “faults on cable A (rate 0.4/m) plus faults on cable B (rate 0.7/m) over a 5 m length of each”, where you need $X \sim \text{Po}(5 \times 0.4 + 5 \times 0.7) = \text{Po}(5.5)$.

Concept: Poisson approximation to the Binomial distribution

When $X \sim B(n, p)$ with **n large** (typically $n > 50$) and **p small** (typically $p < 0.1$, so that $np < 5$ or so), X can be approximated by $Y \sim \text{Po}(np)$. This is useful because Binomial probabilities with large n are impractical to compute by hand, while Poisson probabilities are not.

Why does this work? Imagine chopping the Binomial's n trials into a huge number of tiny sub-intervals. Each sub-interval has a very small chance p of “success”, and there are lots of them (n

large), but the *total* expected number of successes np stays fixed at some moderate value λ . As n grows and p shrinks while $np = \lambda$ is held constant, the Binomial formula ${}^nC_x p^x (1-p)^{n-x}$ converges exactly to $e^{-\lambda} \lambda^x / x!$ — you don't need to reproduce this limit in an exam, but it explains why the two distributions look so similar under “many trials, rare event” conditions (radioactive decay, typos, road accidents), and why matching $\lambda = np$ is exactly the right thing to do.

Worked example 3: Poisson approximation to Binomial

EXAMPLE 2.3

Question. A component is defective with probability 0.008. A batch contains 400 components. Use a suitable approximation to find the probability that at most 2 components in the batch are defective.

Step 1. Recognise that $X \sim B(400, 0.008)$ has large n and small p , with $np = 400 \times 0.008 = 3.2$, so approximate with $Y \sim \text{Po}(3.2)$. **(M1)**

Step 2. $P(Y \leq 2) = P(Y=0) + P(Y=1) + P(Y=2) = e^{-3.2} + e^{-3.2}(3.2) + e^{-3.2}(3.2)^2/2 = e^{-3.2}(1 + 3.2 + 5.12) = e^{-3.2}(9.32) \approx 0.3799$. **(M1A1)**

Answer: 0.380 (3 s.f.)

EXAM TIP

Always state np explicitly and justify “ n large, p small” in words before using the approximation — exam mark schemes frequently award a method mark specifically for stating the correct approximating parameter, and it separates understanding the approximation from having memorised “just use Poisson with mean np ” without knowing why it is valid.

Chapter summary

Idea	Key point
Conditions for Poisson	Events occur singly, randomly, independently, at a constant average rate
$P(X = x)$	$e^{-\lambda} \lambda^x / x!$
Mean and variance	$E(X) = \text{Var}(X) = \lambda$
Sum of independent Poissons	$X \sim \text{Po}(\lambda), Y \sim \text{Po}(\mu)$ independent $\Rightarrow X + Y \sim \text{Po}(\lambda + \mu)$
Poisson approximation to Binomial	Valid when n is large (typically $n > 50$) and p is small (typically $p < 0.1$); use $\text{Po}(np)$

Common exam mistakes

- Forgetting that λ must be scaled when the interval changes (e.g. rate per minute but a question about a 5-minute interval).
- Using $P(X = 0)$ as “e to the power lambda” instead of e to the power *negative* lambda.

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- Trying to sum an infinite series directly for “at least” questions instead of using the complement.
 - For “at least 2” (rather than “at least 1”), forgetting that the complement of $\{X \geq 2\}$ is $\{X = 0 \text{ or } X = 1\}$, not just $\{X = 0\}$ — both terms must be subtracted.
 - Quoting the Poisson approximation to Binomial without checking (or stating) that n is large and p is small.
 - Forgetting that $\text{Var}(X) = \lambda$ too, not just $E(X) = \lambda$, when a question asks for the standard deviation.

You should now be able to

- State the conditions needed for a Poisson model to be appropriate.
- Calculate Poisson probabilities using $P(X = x) = e^{-\lambda} \lambda^x / x!$, including “at least” and “at most” probabilities.
- Combine independent Poisson variables by adding their means, including scaling λ to a different interval length.
- Apply the Poisson approximation to the Binomial distribution when n is large and p is small, and justify why the approximation is being used.